Jilu P Jose

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Projects

W.I.K.T.O.R Chatbot | *GitHub* **Dec '25 – Present**

- Engineered an intelligent RAG-based chatbot designed to deliver accurate, real-time responses using Wikipedia as a dynamic knowledge source.
- Architected the full RAG pipeline using Gemini API, integrated vector embeddings, and reduced Wikipedia API retrieval time by 30%.
- Achieved 95% generation accuracy and MSE of 0.04 while enabling context-aware, up-to-date information delivery across diverse user queries.

Tech: *Python, Streamlit, Wikipedia API, Hugging Face*

S.E.N.T.R.A System | *GitHub* | *Live* **Dec '25 – Jan '26**

- Optimized a real-time facial-emotion recognition system using CNN-based model, achieving 80% classification accuracy across 7 core emotion categories with live webcam.
- Designed an automated logging pipeline with SQLite integration, capturing 100% timestamped emotion events and reduced manual HR review effort by 40%.
- Integrated a Decision Tree Classifier model to recommend personalized activities for detected negative emotions, improving intervention accuracy by 30%.

Tech: *Python, OpenCV, TensorFlow/Keras, SQLite, HTML5, CSS3, JavaScript*

N.O.V.A System | *GitHub* | *Live* **Oct '25 – Dec '25**

- Spearheaded an end-to-end satellite health monitoring system using simulated telemetry data collected from an Arduino-based multi-sensor platform, achieving 90% sensor data reliability.
- Programmed a multi-class machine learning framework incorporating 8 distinct classification algorithms achieving 20% dimensionality reduction for improved computational efficiency.
- Attained an average classification accuracy of 91% with a Mean Squared Error (MSE) of 0.05, and deployed the solution through an interactive web application using Flask as the backend.

Tech: *Python, HTML5, CSS3, JavaScript, Flask*

Training

Lovely Professional University **Jun '25 – Jul '25**

From Data to Decisions | GitHub | Certificate

- Drafted and implemented a multiple regression model to interpret complex environmental variables enabling data-driven insights into annual crop production trends.
- Enhanced the model's predictive performance from 67% to 96% significantly thereby making the solution suitable for real-world forecasting and decision making.
- Performed data preprocessing by reducing dimensionality by 30% to lower computation cost and handled outliers using the IQR (Inter-Quartile Range) method.

Tech: *MySQL, Python, Power BI, Machine Learning, Microsoft Excel*

Certifications

- Oracle Database @AWS Certified Architect Professional | ***Oracle*** **Oct '25**
- Oracle Cloud Infrastructure 2025 Certified DevOps Professional | ***Oracle*** **Oct '25**
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | ***Oracle*** **Sep '25**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | ***Oracle*** **Sep '25**

Technical Skills

- **AI/ML/DL:** Regression, Classification, Clustering, Neural Networks, CNN, RNN, RAG
- **Languages:** Java, C++, C, Python, JavaScript
- **Frameworks:** TensorFlow/Keras, NumPy, Pandas, Polars, Scikit-Learn, Seaborn, Matplotlib, Flask
- **Tools/Platforms:** VSCode, Jupyter Notebook, Power BI, N8N, Automation Anywhere
- **Database:** MySQL, MongoDB, PostgreSQL

Education

Lovely Professional University Punjab **Aug '23 – Present**

*BTech in Computer Science and Engineering — **CGPA: 8.1*** *Jalandhar, Punjab*

Kendriya Vidyalaya Dimapur **Apr '21 – Apr '22**

*12th Science (CBSE) - **76%*** *Dimapur, Nagaland*

Kendriya Vidyalaya 3BRD Air Force Station Chandigarh **Apr '19 – Apr '20**

*10th (CBSE) - **75%*** *Chandigarh, Punjab*